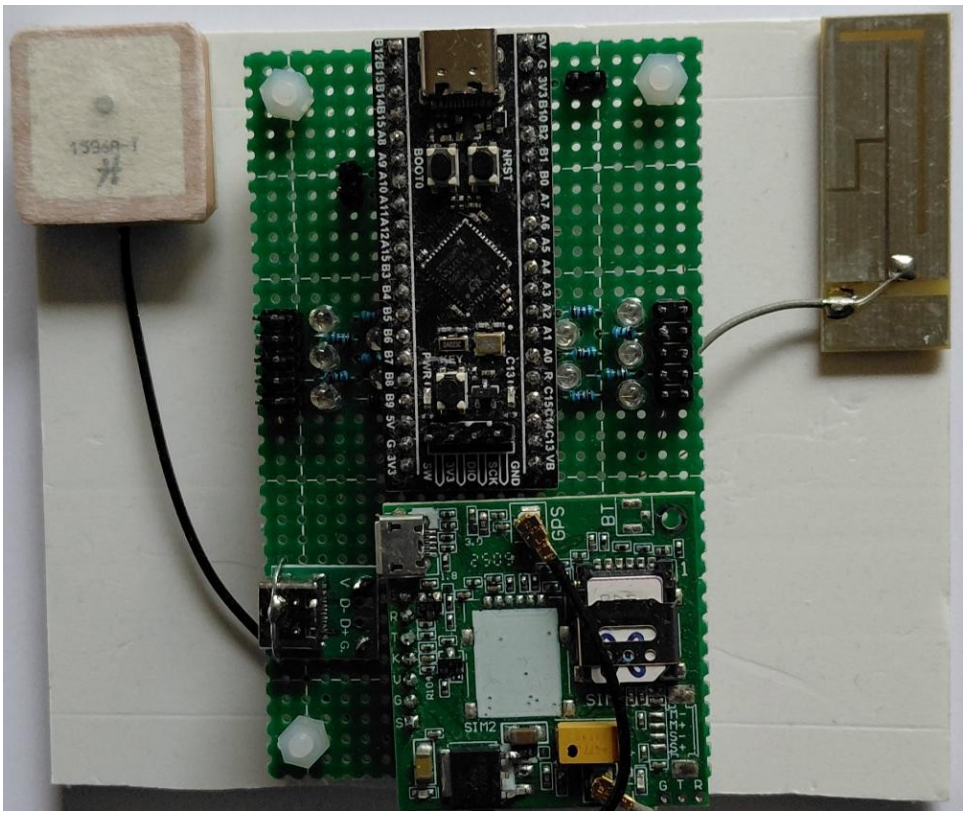
SMS Tracker met enkele extra’s



Built using an A7670E 4G SIM module and an STM32F411 board.

Functionality:

Reports the module's location via SMS.

Includes 5 inputs and 5 outputs that can be queried or controlled via SMS.

Provides location notifications in the following scenarios:

When the module is powered on.

When the status of any of the 5 inputs changes.

If the module remains powered on:

Sends a notification if the module is moved after a configurable time period (e.g., for a parked vehicle).

Upon receiving a specific SMS query.

Upon restart, outputs can be set to a default state (e.g., <0>) or restored to the state they were in prior to power-down.

list function showing the current status of all inputs/outputs and all other programmable functions (see further details).

Automatic notifications can be disabled.

Automatically switches to 'parked' status if no programmable movement has occurred within a configurable period of time.

Important
The SIM7670E serial port is set to 115200 baud by default.
This does not work for my setup, as the program I wrote requires a baud rate of 9600.

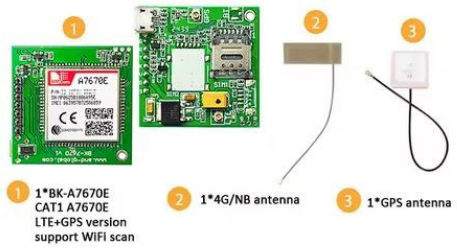
To change this:
Connect the SIM7670E to a terminal (such as the one in the Arduino IDE) using an FTDI module.

Set the terminal to 115200 baud.

Type `AT` <enter>
SIM7670E response: `OK`
Type `AT+IPREX=9600` <enter>
Set the terminal to 9600 baud.
Type `AT` <enter>
SIM7670E response: `OK`

Parts

1x Sim7670E



https://nl.aliexpress.com/item/1005004176439432.html?spm=a2g0o.productlist.main.3.23b97d21Mvrjlv&algo_pvid=be2ed57c-e11b-43c1-b2a3-a82956cb3bf3&algo_exp_id=be2ed57c-e11b-43c1-b2a3-a82956cb3bf3-35&pdp_ext_f=%7B%22order%22%3A%22885%22%2C%22spu_best_type%22%3A%22price%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21EUR%2122.03%2119.57%21%21%2124.12%2121.43%21%4021038c6f17941344419125984e1178%2112000028288768404%21sea%21BE%210%21X%211%210%21n_tag%3A-29919%3Bd%3Aa857d492%3Bm03_new_user%3A-29895%3BpisId%3A5000000221085700&curPageLogUid=DyI4ty415Aqr&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005004176439432%7C_p_origin_prod%3A

1x STM32F411



https://nl.aliexpress.com/item/1005001456186625.html?spm=a2g0o.productlist.main.1.378757e7i42pWC&algo_pvid=4378427b-e13f-4ff5-b74b-b60d57ba9b61&algo_exp_id=4378427b-e13f-4ff5-b74b-b60d57ba9b61-0&pdp_ext_f=%7B%22order%22%3A%227583%22%2C%22spu_best_type%22%3A%22price%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21EUR%215.88%215.69%21%21%2142.91%2141.52%21%402103956a17941346520714099e0ecc%2112000034304092617%21sea%21BE%210%21X%211%210%21n_tag%3A-29919%3Bd%3Aa857d492%3Bm03_new_user%3A-29895%3BpisId%3A5000000221157434&curPageLogUid=MbFkTdTsFOsW&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005001456186625%7C_p_origin_prod%3A

1x SIM card <remove PIN code>
To remove the PIN code, insert the SIM card into a mobile phone and search on Google for instructions on how to do this for your specific device.

5-volt power supply
must be capable of delivering peak currents of at least 2A.
If the connection to the telecom provider drops frequently, the power supply is likely the cause.

5x R 10K

Optional (LEDs on inputs and outputs)

10x led
10x R 5K6

Connections

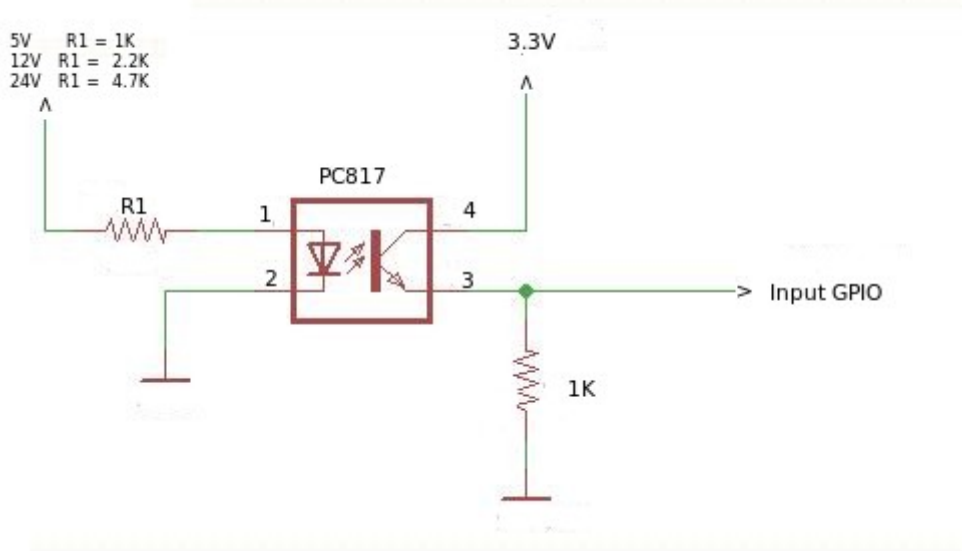
STM32F411		**SIM7670E**
GND	---------- GND Power ----------	G
5V	---------- 5V Power ----------	V
PA3	----------	T
PA2	----------	R
PB5	----- R10K ----- GND	
PB6	----- R10K ----- GND	
PB7	----- R10K ----- GND	
PB8	----- R10K ----- GND	
PB9	----- R10K ----- GND	

Optional LED on input / output

input / output GPIO ----- R5K6 ----- +LED ----- GND

Watch out input / output 3.3 Volt

for higher input voltages



for higher output voltages

IRF520 module
https://nl.aliexpress.com/item/1005006157177189.html?spm=a2g0o.productlist.main.1.1900O7dKO7dKOg&algo_pvid=c5e6c073-2384-417f-9968-9cb927321b1a&algo_exp_id=c5e6c073-2384-417f-9968-9cb927321b1a-0&pdp_ext_f=%7B%22order%22%3A%22531%22%2C%22eval%22%3A%221%22%7D&pdp_npi=4%40dis%21EUR%213.61%211.63%21%21%213.65%211.65%21%40211b81661739390704768899 3ec539%2112000036028414526%21sea%21BE%210%21ABX&curPageLogUid=GH7MeJvjNbDq&utparam-url=scene%3Asearch%7Cquery_from%3A#nav-specification



or

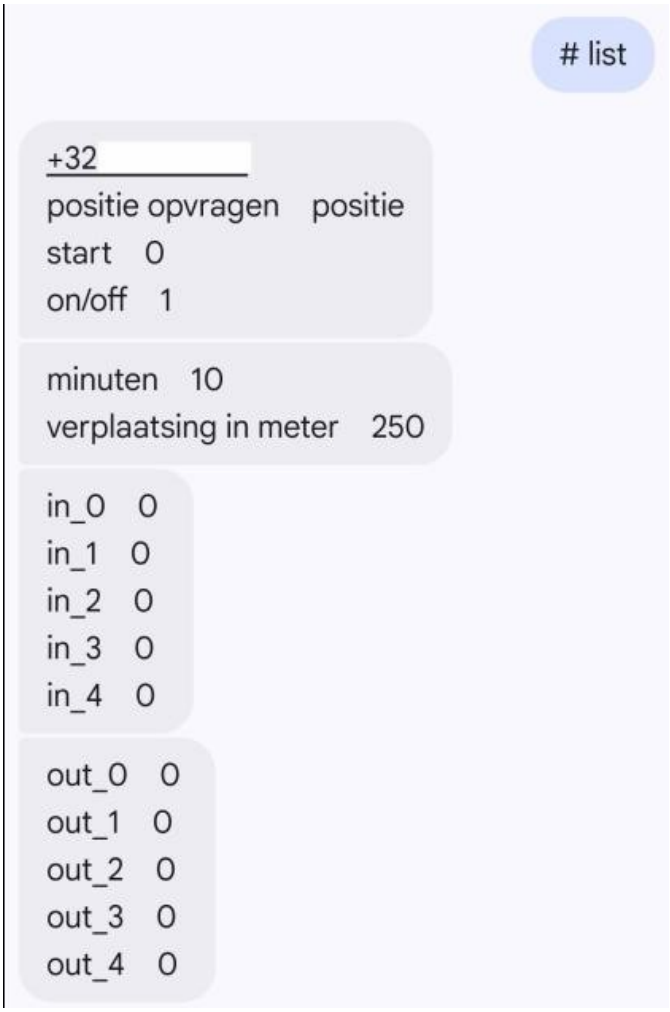
https://nl.aliexpress.com/item/1005003750654499.html?spm=a2g0o.productlist.main.1.34063611FvNnqT&algo_pvid=d6960f06-057f-4841-a1a2-fd148195d4e3&algo_exp_id=d6960f06-057f-4841-a1a2-fd148195d4e3-0&pdp_ext_f=%7B%22order%22%3A%222345%22%2C%22eval%22%3A%221%22%2C%22fromPage%22%3A%22search%22%7D&pdp_npi=6%40dis%21EUR%211.01%211.00%21%21%211.10%211.09%21%400b0fe40d17942377464864321e0d87%212120000270382570 56%21sea%21BE%210%21X%211%210%21n_tag%3A-29919%3Bd%3Af17f58ec%3Bm03_new_user%3A-29895&curPageLogUid=Hg9gAcHHncVB&utparam-url=scene%3Asearch%7Cquery_from%3A%7Cx_object_id%3A1005003750654499%7C_p_origin_prod%3A



The module can be accessed or programmed by sending SMS messages to it.
Each command begins with a # followed by a space.
If the module does not respond to an SMS, there is likely an error in the message sent.
The name of an input/output or the name used in a position-query command must not exceed 12 characters.
Please note: due to a bit of laziness on the part of the programmer, there is no validation for the correctness of the SMS messages sent; please ensure they are correct yourself.

Initial values EEPROM

list
Retrieving configured values as well as current input and output values
In all of the following screenshots, the command sent is shown on the right, with the module's response displayed below it on the left.
Here, the command `# list` is shown, with the response below it.



Commands to adjust initial values

When the module starts up, SMS messages are sent to the number programmed in the EEPROM, provided that no SMS has yet been received.
If the module has received an SMS, the messages are sent to the number associated with the most recently received SMS.

+…….
To change this number in the EEPROM,
replace the number +3200000000 with your mobile phone number. The number must be provided in internationalformat (i.e., starting with +…).



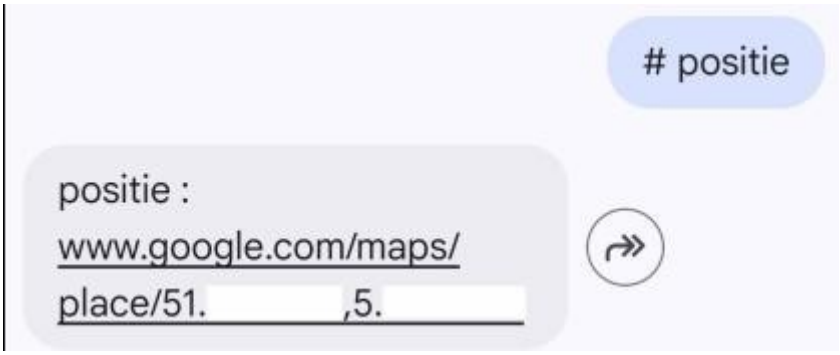
com
To request the position, use the command following <request position> in the listing. You can also change this command; see further details below.

Requesting position if it is not yet available



Initially, it provides a notification that the position is unavailable. Once the position becomes available, a second message is sent automatically.

Request position if position is available



Command for changing the retrieval command



After executing this command, you must request the position using
war ben je

start
value 0 or 1
if 0 outputs are set to 0 upon startup
if 1 outputs are set to the values they held at the time of power failure upon startup



on/off

if 0	no automatically sent SMS messages
if 1	automatically sent messages enabled

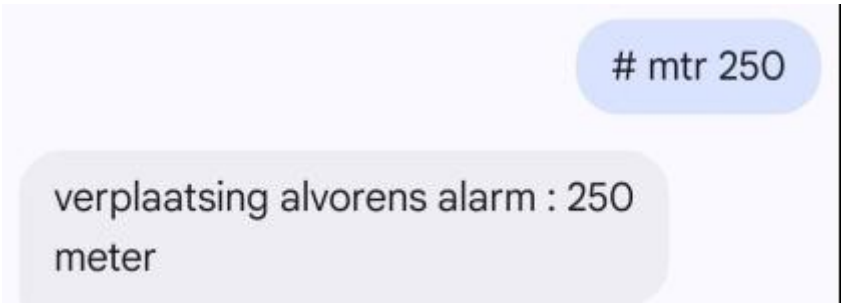
These messages are sent automatically if 'on/off' is set to 1:
Module startup message
Change in input value
Vehicle parked
Vehicle movement greater than the specified threshold after being parked



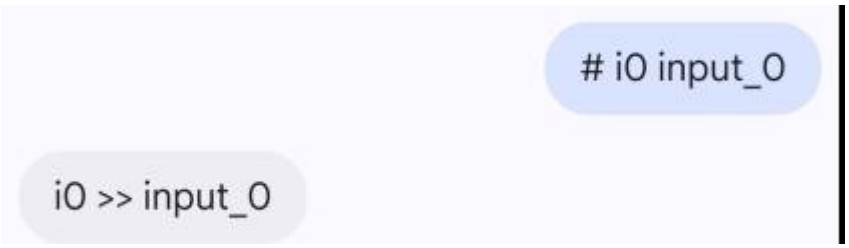
min
Enter the duration for which the vehicle must remain stationary (i.e., move no more than the specified distance in meters—see below) before a "parked" notification is triggered.



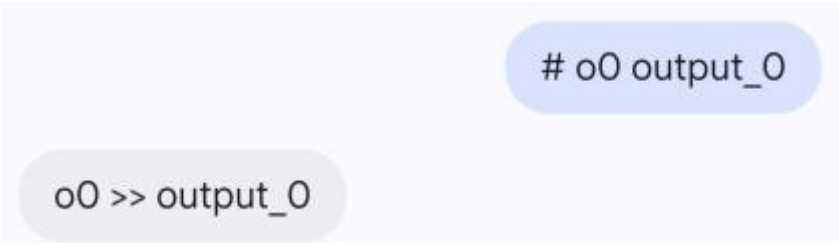
mtr
Enter the distance in meters that the vehicle is allowed to move before an alert is triggered. To avoid false alerts, it is recommended to set the distance to no less than 250 meters.



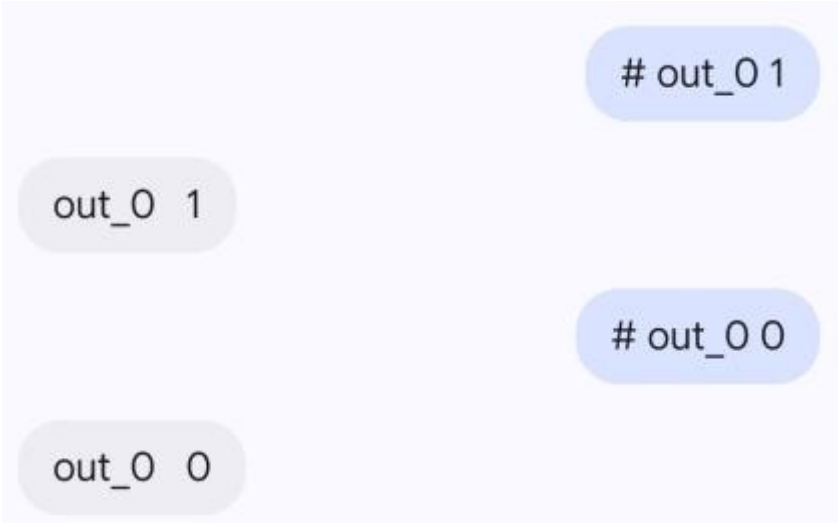
i0 / # i1 / # i2 / # i3 / # i4
Changing the name of inputs. This is used when reporting a change in an input value.
Changing the name of i0 here.



o0 / # 01 / # o2 / # o3 / # o4
Changing the name of the outputs; this name must be used in order to control an output.
Changing the name of o0 here



out_0 / # out_1 / # out_2 / # out_3 / # out_4
controlling outputs
assuming that the names of the outputs are still out_0 ….. out_4; otherwise, you will need to change these to the names you have chosen.
Here, controlling out_0



The report on the status of the output represents the actual current value of the output.
digitalRead()

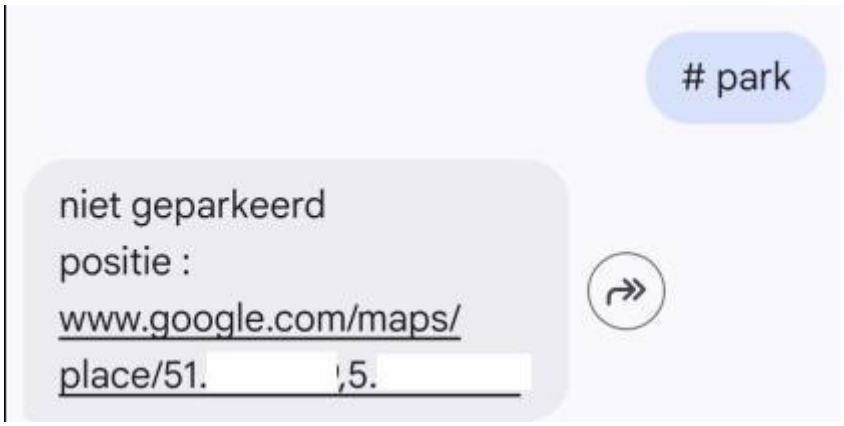
set
sends all outputs <1>

rst
controls all outputs <0>



park
check whether the vehicle is parked

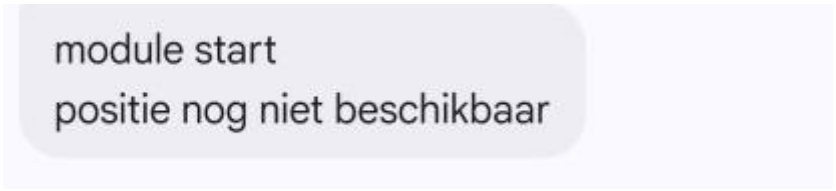




more about parking, see below

Automatic notifications

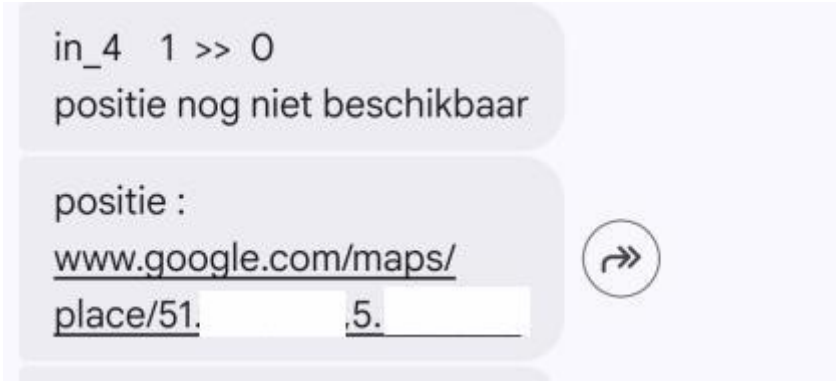
Startup message



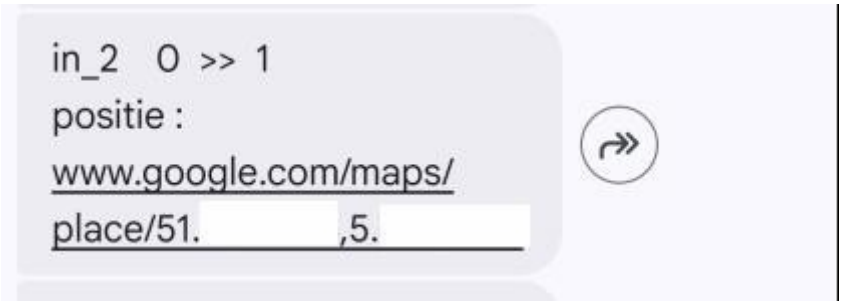
At startup, no position is available yet; once a position becomes available, a second message is sent automatically.



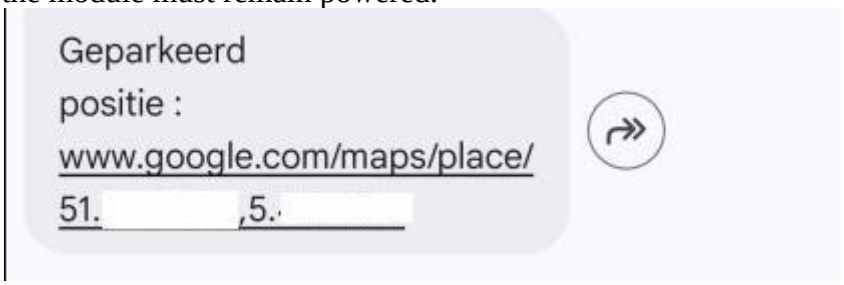
Changing input status
If the position is not yet available, a second message will follow automatically once it becomes available.



position is available



Parked
For this to work, the module must remain powered.



Displacement of more than … meters
Displacement set to 2 meters, so I don't have to jump on my bike and travel more than 250 meters just to trigger a notification.



Well, that's it—enjoy!

regards,
thieu-b55